

Marcelo Lares | Currículum Vitae

Villa Carlos Paz – Córdoba – Argentina

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Overview

Experience	p.1	Applied AI technical leadership, production ML, and research 20 years teaching, 10+ years research, 4+ years industry
Current position	p.3	Associate Professor at Universidad Nacional de Córdoba Member of AAA, IAU, TOROS & OTHER
Skills	p.3	production ML, experiment strategy, statistics, astrostatistics
Education	p.4	PhD in Astronomy, Argentina
Publications	p.4	30+ articles in international Q1 journals, including large collaborations 28 articles in proceedings 7 software packages (2 indexed), 3 public catalogues 17 <i>h</i> -index, excluding large collaborations
Teaching	p.6	20 years of teaching experience
Mentoring	p.7	2 PhD students, 1 finished master degree student
Evaluation	p.7	Reviewer for 2 journals and 1 meeting, 2 PhD Theses
Service	p.7	6 years as technical editor, LOC for 2 meetings
Grants	p.8	member of 23 grants (PI on 5 grants)
Outreach	p.8	speaker at 3 conferences, 12 media mentions 2 industry services, 1 COVID19 service
Meetings	p.10	10 international meetings, 17 national meetings speaker at 6 meetings, 1 invited talk

Experience

I am an Applied AI and Data Science technical leader with experience in production recommendation systems, client-facing AI delivery, experiment strategy, cross-functional execution, research, and university-level teaching. I combine model ownership with roadmap thinking, formal validation, mentoring, a PhD in Astronomy, 30+ articles in international Q1 journals, and extensive teaching and supervision experience. In industry, I have worked on LATAM-scale recommendation systems, production ranking models, A/B-test decision support, feature-store integration, MLflow experiment tracking, ONNX release validation, and client-facing AI products. I have led an eight-person data science team, defined technical roadmaps and OKRs, mentored data scientists, and translated business requirements into data products in e-commerce, public revenue, retail, tourism, agriculture, and international trade. As a researcher, I took part in research projects and teams to study topics in astronomy, astrostatistics, cosmology, transient events, and scientific software. I was principal investigator in several projects. My publication list includes papers from international and interdisciplinary collaborations. I also participated in and organized scientific conferences, and was the technical editor of a scientific journal of national scope for 6 years. As a teacher, I have developed communication skills and was responsible for courses on mathematics, physics, informatics, astronomy and statistics. I also have extensive experience in public outreach. I consider myself a curious researcher, and enjoy learning new technologies. I am interested in the effective communication of ideas and methods through visualization and storytelling.

Recent professional experience.....

Mercado Libre

Senior Data Scientist / Machine Learning Engineer, Argentina

2024–Jun 2026

- Served as the primary data science owner for a production online ranking model powering recommendations for dozens of internal clients and millions of users across Argentina, Brazil, and Mexico.
- Guided production experimentation decisions for ranking models, recommending when to launch, pause, or refine experiments based on statistical evidence, business KPIs, cost, and operational risk.
- Developed and maintained the PyTorch ranking model, adding features, modifying embeddings, and integrating production feature-store data.
- Adapted Fury pipelines and optimized framework and CUDA configurations, tripling the amount of training data the model could process.
- Prepared ONNX artifacts, validated production configurations, and partnered with engineering to ensure reliable production deployments.
- Implemented MLflow experiment tracking and designed a pre-production validation protocol for safer, reproducible model releases.
- Developed statistical tools to quantify confidence in A/B-test differences and monitored clicks, purchases, GMV, and model behavior through Datadog and Looker.
- Communicated trade-offs, risks, and next steps to engineering, product, analytics, and leadership stakeholders, shaping model-release decisions.
- Delivered measurable lift in purchase conversion through controlled production experiments while coordinating engineering, product, and analytics teams.

IThreex Global

Lead Data Scientist, Argentina

2022–2024

- Led an eight-person data science team delivering client-facing AI and analytics solutions across public revenue, retail, tourism, agriculture, and international trade.
- Defined the technical roadmap for the Molibdeno AI platform, prioritizing reusable workflows, client needs, and delivery constraints.
- Defined OKRs and tracked team execution toward client-facing AI and analytics deliverables.
- Designed and implemented the core Python library used by the Molibdeno AI platform to standardize reusable machine-learning workflows.
- Served as the principal technical contact for Kolektor, developing payment-behavior analyses and tax-revenue forecasts for large taxpayers in Córdoba.
- Delivered segmentation and predictive solutions that supported marketing and commercial strategies for Pueblo Nativo, Inverfin, and Agencia ProCórdoba.
- Developed computer-vision models for image-based cattle-weight estimation and defined the corresponding image-acquisition protocols.
- Built a LangChain-based RAG support service exposed through an API.
- Presented technical progress and results to clients and stakeholders, translating business requirements into data products.
- Led, mentored, and onboarded data scientists, assigning technical direction across client projects.
- Participated in candidate interviews, technical hiring decisions, performance evaluation, and difficult staffing decisions.

Current position

- Associate Professor at Universidad Nacional de Córdoba, Argentina.
- Former Senior Data Scientist / Machine Learning Engineer at Mercado Libre, Argentina (ended June 2026).
- Former CONICET researcher, Argentina.
- Member of the IAU, the AAA (Argentine Astronomical Association), the TOROS project and the OTHER project.

Technical skills

- Programming languages and environments: Python, R, Fortran, SQL, bash, Linux.
- Languages: Spanish: Native – English: Advanced – Portuguese: Basic.
- Machine learning: PyTorch, Scikit-Learn, XGBoost, ranking models, recommender systems, embeddings, feature engineering, computer vision, supervised and unsupervised learning.
- Production ML and experimentation: MLflow, ONNX, feature stores, Fury pipelines, CUDA configuration, A/B testing, pre-production validation, Datadog, Looker.
- Data science and statistics: NumPy, Pandas, SciPy, Matplotlib, Bayesian inference, statistical learning, Monte Carlo methods, confidence analysis, model validation.
- Data, cloud and APIs: BigQuery, SQLite, AWS S3/EC2, GCP BigQuery and Cloud Storage, REST APIs, LangChain, retrieval-augmented generation.
- Scientific and data engineering: high performance computing, parallel programming, OpenMP, reproducible pipelines, data-intensive workflow design, Sphinx, Read the Docs.
- Software development: git, GitHub, GitLab, continuous integration, testing, documentation, HTML, Jekyll, WordPress.
- Leadership and delivery: technical roadmaps, OKRs, mentoring, onboarding, hiring participation, client communication, stakeholder management, agile delivery.
- Reporting and communication: $\text{\LaTeX}$, scientific writing, technical presentations, dashboards, data visualization and public communication.
- More details on [my LinkedIn profile](#).

Education

- **Ph.D. in Astronomy**
○ *National University of Córdoba, Argentina* 2003–2009
Supervisor: Prof. Dr. Diego García Lambas
- **National University of Córdoba (UNC)**
○ *Licenciado degree in Astronomy, Argentina* 1997–2003
(The Licenciado Degree comprises 32 courses including a research project in astronomy in a minimum of five years)
- **Instituto "El Obrero"**
○ *Técnico químico, Alta Gracia–Córdoba* 1991–1996
Best score, flag escort

Publications

- PhD Thesis: Low luminosity galaxy satellites
- 30+ articles in international Q1 journals
- Articles with large scientific collaborations
- 28 refereed articles on Conferences
- 6 unpublished presentations in conferences
- 7 software packages and 3 public data catalogues

An updated list of my publications can be found in my ORCID profile:

<https://orcid.org/0000-0001-8180-5780>

Presentations on Conferences (not published)

- [1] **PISCIS: A Platform for Interactive Search and Citizen Science**
2020, In *Reunión Anual de la Asoc. Argentina de Astronomía (accepted, in press)*
I.V. Daza, I.G. Alfaro, J.A. Benavides, M. Lares, M. Santucho, J.B. Cabral, A.L. O'Mill, F. Rodriguez, and M. Koraj.
- [2] **Soporte de decisiones frente al COVID-19**
2020, In *Reunión Anual de la Asoc. Argentina de Astronomía (accepted, in press)*
M. Lares, J.B. Cabral, I.V. Daza, N. Luczywo, D. Paz, B. Sanchez, and F. Stasyszyn.
- [3] **The role of the galactocentric distance of a civilization on its chance of contacting other intelligent civilizations**
2020, In *Reunión Anual de la Asoc. Argentina de Astronomía*
Marcelo Lares, José Funes, and Luciana Gramajo.
- [4] **Gravitational-Wave Optical Counterpart Detection Methods for the TOROS Campaign During LVC O2 Observation Run**
2019, In *APS April Meeting Abstracts*
Martin Beroiz, Richard Camuccio, Moises Castillo, Mario Diaz, Mariano Dominguez, Marcelo Lares, Bruno Sanchez, Adam Zdrozny, and Toros Team Transient Optical Robotic Observatory Of The South.
- [5] **The corral framework for pipelines**

2017, In *Astronomical Data Analysis Software & Systems (Conferencia)*

Bruno Sanchez, Juan Bautista Cabral, Martín Beroiz, Mariano Dominguez, Marcelo Lares, and Sebastián Gurovich.

Technical developments.....

[1] **AEGIS: Academic Exam Generator for Interchange and Shuffle**

2020, page <https://github.com/mlares/aegis>

Marcelo Lares.

[2] **Docker container for PHOEBE**

2020, page <https://hub.docker.com/r/arunalum/phoebe>

Marcelo Lares.

[3] **PINNACLE: Publications from an Institute: Numbers, Networks, Authors and Citations Looking for Excellence**

2020, page <https://github.com/mlares/pinnacle>

Marcelo Lares.

[4] **HEARSAY: Simulations for the probability of alien contact**

2020, pages ascl:2006.001, <https://ascl.net/2006.001>

Marcelo Lares, José Funes, Luciana Gramajo, and Juan Cabral.

→ Reviewed and indexed by Astrophysics Source Code Library

[5] **GriSPy: Fixed-radius nearest neighbors grid search in Python**

2019, pages ascl:1912.013, <https://ascl.net/1912.013>

Martín Chalela, Emanuel Sillero, Luis Pereyra, Mario A. Garcia, Juan B. Cabral, Marcelo Lares, and Manuel Merchán.

→ Reviewed and indexed by Astrophysics Source Code Library

[6] **Properimage: Image coaddition and subtraction**

2019, pages ascl:1904.025, <http://ascl.net/1904.025>

Bruno O. Sánchez, Juan B. Cabral, M. Beroiz, M. Domínguez, and M. Lares.

→ Reviewed and indexed by Astrophysics Source Code Library

[7] **CPF: Corral Pipeline Framework**

2018, pages ascl:1808.003, <https://ascl.net/>

Juan Cabral, Bruno Sanchez, Martin Beroiz, Mariano Dominguez, Marcelo Lares, Sebastian Gurovich, and Pablo Granitto.

→ Reviewed and indexed by Astrophysics Source Code Library

[8] **VizieR Online Data Catalog: Opt. follow-up of GW170817 counterpart (Diaz+, 2017)**

2018, *VizieR Online Data Catalog*, page J/ApJ/848/L29

M. C. Diaz, L. M. Macri, D. Garcia Lambas, C. Mendes de Oliveira, J. L. Nilo Castellon, T. Ribeiro, B. Sanchez, W. Schoenell, L. R. Abramo, S. Akas, J. S. Alcaniz, R. Artola, M. Beroiz, S. Bonoli, J. Cabral, R. Camuccio, M. Castillo, V. Chavushyan, P. Coelho, C. Colazo, M. V. Costa-Duarte, H. Cuevas

Larenas, D. L. Depoy, M. Dominguez Romero, D. Dultzin, D. Fernandez, J. Garcia, C. Girardini, D. R. Goncalves, T. S. Goncalves, S. Gurovich, Y. Jimenez-Teja, A. Kanaan, M. Lares, R. Lopes de Oliveira, O. Lopez-Cruz, J. L. Marshall, R. Melia, A. Molino, N. Padilla, T. Penuela, V. M. Placco, C. Quinones, A. Ramirez Rivera, V. Renzi, L. Riguccini, E. Rios-Lopez, H. Rodriguez, L. Sampedro, M. Schneider, L. Sodre, M. Starck, S. Torres-Flores, M. Tornatore, and A. Zdrozny.

→ Member of the TOROS project

[9] **VizieR Online Data Catalog: Opt. follow-up of GW170817 counterpart (Diaz+, 2017)**

2018, *VizieR On-line Data Catalog: J/ApJ/848/L29*. Originally published in: 2017ApJ...848L..29D, page J/ApJ/848/L29

M C Diaz, L M Macri, D Garcia Lambas, C Mendes de Oliveira, J L Nilo Castellon, T Ribeiro, B Sanchez, W Schoenell, L R Abramo, S Akas, J S Alcaniz, R Artola, M Beroiz, S Bonoli, J Cabral, R Camuccio, M Castillo, V Chavushyan, P Coelho, C Colazo, M V Costa-Duarte, H Cuevas Larenas, D L Depoy, M Dominguez Romero, D Dultzin, D Fernandez, J Garcia, C Girardini, D R Goncalves, T S Goncalves, S Gurovich, Y Jimenez-Teja, A Kanaan, M Lares, R Lopes de Oliveira, O Lopez-Cruz, J L Marshall, R Melia, A Molino, N Padilla, T Penuela, V M Placco, C Quinones, A Ramirez Rivera, V Renzi, L Riguccini, E Rios-Lopez, H Rodriguez, L Sampedro, M Schneider, L Sodre, M Starck, S Torres-Flores, M Tornatore, and A Zdrozny.

[10] **VizieR Online Data Catalog: LMC star clusters catalog (Palma+, 2016)**

2016, *VizieR On-line Data Catalog: J/A+A/586/A41*. Originally published in: 2016A&A...586A..41P, pages J/A+A/586/A41

T Palma, L V Gramajo, J J Claria, M Lares, D Geisler, and A V Ahumada.

An updated sample of software projects can be found in my GitHub profile:

<https://github.com/mlares>

Teaching Activities

Teaching positions.....

○ OAC Associated Professor, Córdoba Argentina	FAMAF-UNC 2023 – today
○ OAC Adjoint Professor, Córdoba Argentina	FAMAF-UNC 2017 – 2023
○ OAC Assistant Professor, Córdoba Argentina	FAMAF-UNC 2007 – 2017
○ FaMAF Teaching assistant, Córdoba Argentina	UNC 2001 – 2006

Mentoring

Students.....

- **Dávila Kurbán, Federico** **UNC**
Ph.D. program, Facultad de Matemática, Astronomía y Física 2017–2022
The internal structure of cosmic voids
- **Daza Perilla, Vanessa** **UNC**
Ph.D. program, Facultad de Matemática, Astronomía y Física 2017–2023
Machine learning detection of transient and periodic objects
- **Luparello, Heliana** **UNC**
graduate program, Facultad de Matemática, Astronomía y Física 2015–2016
The largest structures in the Universe

Evaluation

Peer-review.....

- **reviewer**
Montly Notices of the Royal Astronomical Society 2020
- **reviewer**
Astronomy and Astrophysics 2004
- **reviewer**
BAAA 2017

Scientific organizing Comitees.....

- **FoF2018** **Córdoba**
Friends of friends meeting 2018
<https://www.fof.oac.unc.ar/>

Service

Editorial.....

- **Boletín de la Asociación Argentina de Astronomía** **AAA**
Technical Editor, Argentina 2011-2017
http://www.astronomiaargentina.org.ar/boletin_aaa
- **3rd. stellar astrophysics meeting** **AAA**
Technical Editor, Argentina 2017
<http://jae2016.oac.uncor.edu/>

Local organizing Comitees.....

- **FoF 2018** Córdoba
Friends of Friends meeting (head of the LOC) 2018
<http://fof.oac.uncor.edu/2018/>
- **SATEC 2021** Córdoba
Scientific And Technical Exchange Conferences (head of the LOC) 2021
<https://satec.iate.conicet.unc.edu.ar>

Grants

- Member of 23 research grants
- PI of 5 research grants

Outreach

Science Outreach.....

- **Pint of Science** Invited speaker
Conference, <http://pintofsciencear.com> 2023
- **OAC general public conferences** Speaker
Conference, OAC 2018, 2017, 2015, 2022
- **Night of the museums** Organizer, speaker
Conference, organizer, OAC 2019, 2018, 2017
- **ScientificAmerican** Press
, Want to Talk to Aliens? Try Changing the Technological Channel beyond Radio, by Adam Mann 29/04/2020
- **Nature news** Press
Press, "The sparkling universe: coherent motions of cosmic voids" by Ron Cowen 10/2015
- **OAC Exhibitions** Organizer
Gagarin 50 - FETTU - OAC 140 Aniversary 2010-2012
- **OAC Web page** Developer
design, administration and maintenance, www.oac.uncor.edu 2010-2012
- **BIAAA** Editor
Informative bulletin of the Argentine Astronomy Asociation 2008-2010
- **EABA - OAC** Visitor guide
Summer seasons at astrophysical station, guided tours to schools 2003-2005

Media, interviews.....

- **Como es la tecnología con la que se analizan los escenarios post-cuarentena.** Periferia
Press note 29/04/2020
Argentina

- **Investigadores del OAC proponen herramientas para entender el coronavirus.** **Prensa UNC**
Press Release, Córdoba *11/04/2020*
Argentina
- **Arcovid, el soporte que permite entender el Coronavirus** **El Show del Lagarto. EldoceTV**
TV interview, Córdoba *08/04/2020*
Argentina
- **Comerse la curva.** **Revista El gato y la caja.**
Press note, Mención por Pablo A. González. *05/04/2020*
Argentina
- **Arcovid19: Herramientas para ayudar a entender y combatir el Coronavirus.** **Prensa OAC**
Press release, Córdoba *04/04/2020*
Argentina
- **Noche de los museos 2018** **OAC-UNC**
Public talk, Córdoba *26/10/2018*
Argentina
- **Noche de los museos 2017** **OAC-UNC**
Public talk, Córdoba *01/12/2017*
Argentina
- **TalkRadio**
Radio interview *9/2020*
UK
- **Canal 10**
interview *9/2020*
Argentina
- **Canal 12**
interview *9/2020*
Argentina
- **Catholic news**
mention *9/2020*
- **Radio Nacional Mendoza**
Interview *9/2020*
Argentina
- **Radio de la Universidad Nacional de Mar del Plata**
Interview *9/2020*
Argentina
- **Convenios de cooperación entre la UNSa y proyectos astronómicos en Salta** **General public**
Conference *12/2018*
Argentina
- **En busca de vida extraterrestre inteligente: un singular proyecto cordobés que une ciencia y religión** **General public**
News
Argentina

Community and social services.....

- **Project ARCOVID19** – <https://ivco19.github.io/>

Member of the multidisciplinary group Arcovid19, aimed at designing tools for decision support in the context of the SARS–COVID 19 pandemic. The team also worked on public outreach and participated on several seminars

and meetings about the topic.

○ Industry Outreach

I participated in several collaborations with the industry, including two "high level technological services" certificated by the outreach office of the CONICET.

Meetings

- **CIBSE: XXV IberoAmerican Conference on Software Engineering**
Type: Annual Meetings
Location:
Dates: 13-17 june, 2022
Role: Attendee and [speaker](#).
- **Astronomy Association of Argentina**
Type: Annual Meetings
Location:
Dates: 2000, 2001, 2002, 2003, 2004, 2006, 2007, 2008, 2009, 2010, 2012, 2013, 2014, 2015, 2016, 2017 and 2020.
Role: Attendee, Posters and speaker.
- **COSMOSUR III**
Type: Periodic workshop
Location: Córdoba
Date: 2015
Role: Invited talk
- **Friends of Friends Meetings**
Type: Periodic workshop
Location: Córdoba, Argentina
Date: 2012, 2013, 2014, 2015, 2016, 2017, 2018 (organizer) and 2019.
Role: Attendee, speaker
- **AAA thematic Workshops**
Type: Workshop
Location:
Date: Astronomía Teórica en Argentina: Problemas y Perspectivas (2007), 2011
Role: Attendee
- **SCS PIRE Summer School on the Large Scale Structure-CMB connection**
Type: School
Location: Chile
Date: 2006
Role: Attendee
- **IAG-LENAC XIII Advanced School of Astrophysics & LENAC Special Session**
Type: School
Location: Foz do Iguaçu, Brazil
Date: 2006
Role: Attendee
- **11th Latin-American Regional IAU Meeting**
Type: Periodic Meeting
Location: Pucón, Chile
Date: 2005
Role: Attendee
- **ESO Workshop: Groups of Galaxies in the nearby universe**
Type: Workshop
Location: Santiago, Chile
Date:
Role: Attendee
- **MPA/ ESO/ MPE/ USM Joint Astronomy Conference: Open Questions in Cosmology: the first Billion years**
Type: Conference
Location: Garching, Germany
Date: 2005
Role: Attendee
- **Advanced Chilean School: First Large Scale Structures in the Universe and their evolution**
Type: School
Location: Santiago, Chile
Date: 2004
Role: Attendee
- **Fifth J.J. Giambiagi Winter School of Physics: Precision Cosmology**
Type: School
Location: Buenos Aires, Argentina
Date: 2003
Role: Attendee

Additional information

Several resources about my achievements, production, selected projects, publications, teaching, outreach and supporting documents can be found in my personal website,

<https://mlares.github.io>